3.11 Using hexadecimal numbers:

3.11.1 Add

3.11.2 Subtract

3.11.3 Multiply

Addition

To add hexadecimal numbers,

- Line up the digits by place value
- Add from right to left
- Convert to decimal and add the numbers
    - If a column totals 16 or more, subtract 16, write the remainder as a hex digit, and carry 1 to the next column.
    - If the column totals less than 16, convert the number to a hexadecimal

Examples

$2A_{16} + 1F_{16}$

Solution:

Ones column: A + F = 10 + 15 = 25. Since 25 = 16 + 9, write 9 and carry 1.

Sixteens column: 2 + 1 + 1 (carry) = 4.

Answer: 49_{16}. Check: 42 + 31 = 73, and $49_{16} = 73_{10}$.

$7D_{16} + 8B_{16}$

$3B_{16} + 24_{16}$

$6F_{16} + 2D_{16}$

$217_{16} + 1A5_{16}$

Subtraction

To subtract hexadecimal numbers,

- Line up the digits by place value
- Subtract from right to left
    - If the top digit is smaller, borrow 1 from the next place to the left.
    - Borrowing 1 adds 16 to the current column, while reducing the next column by 1.

Examples

$5C_{16} - 2E_{16}$

Solution:

Ones: C (12) is smaller than E (14). Borrow 1 from the 5, so the ones column becomes 12 + 16 = 28.

Ones: 28 − 14 = 14 = E. Sixteens: 4 − 2 = 2.

Answer: $2E_{16}$. Check: 92 − 46 = 46.

$100_{16} - 1A_{16}$

$8D_{16} - 3A_{16}$

$200_{16} - B7_{16}$

$FC4_{16} - 2C7_{16}$

Multiplication

To multiply hexadecimal numbers,

- Multiply the digits as you would in decimal form.
- Convert each product into a hexadecimal.
    - Divide by 16 to find the carry and the remainder.
- For multi-digit multiplication, shift each partial product one hexadecimal place left as needed.
- Add the partial products in base 16.

Examples

$B_{16} \times 6_{16}$

Solution:

B = 11, so 11 × 6 = 66.

66 = 4 × 16 + 2, so the result is 42_{16}.

$2A_{16} \times 3_{16}$

$1A_{16} \times 12_{16}$

$C_{16} \times 4_{16}$

$3D_{16} \times 2_{16}$

$1B_{16} \times 13_{16}$

$A01_{16} \times 29_{16}$

x	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	10
1	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	10
2	2	4	6	8	A	C	E	10	12	14	16	18	1A	1C	1E	20
3	3	6	9	C	F	12	15	18	1B	1E	21	24	27	2A	2D	30
4	4	8	C	10	14	18	1C	20	24	28	2C	30	34	38	3C	40
5	5	A	F	14	19	1E	23	28	2D	32	37	3C	41	46	4B	50
6	6	C	12	18	1E	24	2A	30	36	3C	42	48	4E	54	5A	60
7	7	E	15	1C	23	2A	31	38	3F	46	4D	54	5B	62	69	70
8	8	10	18	20	28	30	38	40	48	50	58	60	68	70	78	80
9	9	12	1B	24	2D	36	3F	48	51	5A	63	6C	75	7E	87	90
A	A	14	1E	28	32	3C	46	50	5A	64	6E	78	82	8C	96	A0
B	B	16	21	2C	37	42	4D	58	63	6E	79	84	8F	9A	A5	B0
C	C	18	24	30	3C	48	54	60	6C	78	84	90	9C	A8	B4	C0
D	D	1A	27	34	41	4E	5B	68	75	82	8F	9C	A9	B6	C3	D0
E	E	1C	2A	38	46	54	62	70	7E	8C	9A	A8	B6	C4	D2	E0
F	F	1E	2D	3C	4B	5A	69	78	87	96	A5	B4	C3	D2	E1	F0
10	10	20	30	40	50	60	70	80	90	A0	B0	C0	D0	E0	F0	100